PORTFOLIO PROJECT · DATA PIPELINE · MARKET ANALYTICS

Northeast Heating & Cooling Demand Map

Where and when the Northeast needs heating oil, propane and AC service, estimated county by county and month by month from public data

COMPLETE · 4 DASHBOARDS LIVE

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Python · SQL · Tableau

WHY THIS PROJECT

Fuel dealers run a seasonal business with two peaks

Winter

`heating oil · propane`

Fuel delivery volume spikes with cold weather. The Northeast is the heating oil capital of the U.S., and propane is common in rural areas.

Summer

`AC install · repair`

Fuel slows down, but full-service dealers keep crews busy installing and repairing air conditioning.

The gap

`no public dataset`

Nobody publishes monthly fuel demand by county. Software vendors serving these dealers have to guess where the demand is.

Goal: build that view from public data, so dispatch, routing and billing platforms can see where demand is, how it moves through the year, and which dealers fit best.

SCOPE

Every county in the nine Northeast states, every month

9

states: CT, MA, RI, VT, NH, ME, NY, NJ and PA

218

counties, each with modeled monthly demand

3

demand types: heating oil, propane and an AC cooling index

5

recent heating seasons at monthly grain

Plus a Connecticut pilot: a dealer layer that maps fuel dealers against county demand and ranks them by fit for a full-service software platform.

PIPELINE

Nine sources, one schema, one dashboard

Sources

Census · EIA · NOAA

Heating fuel by home, usage, weather, prices and dealers.

Extract

requests · BeautifulSoup

API pulls, file downloads and a dealer scrape, one script per source.

Model

pandas · geopandas

Standardize FIPS codes, convert units, estimate demand and calibrate.

Store + analyze

SQL

Star schema, nine analysis queries and Tableau-ready views.

Visualize

Tableau Public

County maps with a month slider, rankings and KPIs.

Sources: Census ACS and TIGER/Line · EIA RECS, SEDS, SHOPP and 861M · NOAA degree days

DEMAND MODEL

Homes × usage, shaped by the weather

```
est_gallons(county, fuel, month) =
  homes(county, fuel)
  × [ gal_per_hdd(state, fuel) × HDD(county, month)
    + base_gal(state, fuel) / 12 ]
```

Homes

Occupied homes heating with oil or propane, from Census ACS.

Usage

RECS heating gallons per home, divided by 2020 degree days, plus a steady base.

Weather

Each county's own NOAA degree days, so cold winters burn more.

Checks

Priced with EIA SHOPP, then calibrated against EIA SEDS state totals.

DELIVERABLES

From a database to answers a sales team can use

Star schema

SQLite · 5 dimensions · 8 facts

- About 140,000 rows, checked for broken links on load
- Demand, cooling, weather, prices and EIA totals

9 SQL analyses

CTEs · window functions · all 9 built

- Top counties, peak months, winters vs. normal
- Price volatility and who heats with what
- Dealer density and fit score (CT)

Tableau dashboards

Tableau Public · live

- The Thermal Year: county map with a month slider
- Size vs. intensity: year-round opportunity
- Who to call first: Connecticut dealer prospects

Year-round score: 60% heating spend percentile + 40% cooling index percentile, to find the best markets for full-service dealers. Tested: the weights barely change the ranking.

EARLY RESULTS · 2025-26 SEASON

A $16.8 billion market that peaks every January

$16.8B

estimated residential heating oil and propane spend

3.2B

gallons of heating oil, plus 1.0B gallons of propane

Jan

peak month in all 218 counties; Dec to Feb is half the year

$686M

largest county market: Suffolk County, NY

Size vs. intensity: ranked on totals, the best year-round markets are the big NYC and Boston counties. Per home, it is oil-heavy areas with warm summers: Sussex NJ, Northeastern CT, Putnam NY.

CONNECTICUT PILOT · 468 LICENSED DEALERS

Who to call first

468

licensed heating fuel dealers, from the state license registry

20%

also hold an HVAC contractor license: full-service

2,206

oil and propane homes per dealer in Western CT, vs. 922 in the Lower River Valley

#1

Densmore Oil, Stonington: full-service, 22 years licensed, under-served region

Fit score: 50% offers HVAC service + 30% how under-served its region is + 20% years licensed. Built in SQL with window functions, ranked with ROW_NUMBER().

DELIVERY LOGISTICS · DENSMORE OIL, STONINGTON CT · 2025-26

Same deliveries, a third fewer miles

34%

fewer truck miles: 26,909 optimized vs. 40,751 sequenced by hand

4,493

K-factor auto-deliveries to about 1,048 modeled accounts, with real daily weather

163

miles on the coldest day (Feb 8, 7.3°F), vs. 256 by hand, same 2 trucks

2

trucks cover the whole winter with zero run-outs. With 1, tanks run dry from Jan 4

How: Huff market share from the CT dealer registry, degree-day forecast per tank, and daily routing on real road distances (OSRM) with OR-Tools: the route optimization dispatch software leaves to the dispatcher.

STATUS

All eight phases are done

01 · DONE

Planning

Questions, sources, method and schema documented

02 · DONE

Data collection

Census, NOAA and EIA pulls in Python

03 · DONE

Modeling

Demand model, calibration and sensitivity

04 · DONE

SQL

Star schema, load and nine analyses

05 · DONE

Tableau

Four dashboards live on Tableau Public

06 · DONE

CT dealers

Dealer list, fit score and prospect ranking

07 · DONE

Findings

Write-up and dashboard screenshots

08 · DONE

Logistics

One dealer's winter, routed with OR-Tools

Honest limits: county demand is modeled from state averages and weather, not measured. It shows patterns and relative size well, not exact gallons.